PARAMETRIC VECTOR FORM

Four equations, one vector sum

Every solution set of a consistent linear system is one particular solution shifted by the whole null space.

BEFORE

Read off the RREF

$$x_1 = 5 - 2s + t$$
$$x_2 = s$$
$$x_3 = -2 - 3t$$
$$x_4 = t$$

Free variables $x_2 = s$, $x_4 = t$

SPLIT BY PARAMETER

one vector each for the constants, s and t

KEEP THE SIGNS

AFTER

Parametric vector form

$$\begin{pmatrix} 5 \\ 0 \\ -2 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ -3 \\ 1 \end{pmatrix}$$

x = particular solution plus null-space directions

WHAT CHANGED

Four scalar equations became one vector equation, read row by row: each vector holds

WHAT STAYED

The solution set: a 2-dimensional affine subspace of $\mathbb{R}^4$, the null space shifted by the

one kind of term.

particular solution.